

PART-AName of the Student: MIHIR HITESH BIRANISignature of the Student: Mihir Roll No.: 230123035Signature of the Invigilator: Gourav Ku Mahy**INSTRUCTIONS**

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	4	9	18

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

To get the LSEs of θ_1 and θ_2 , we need to ~~minimize~~ ^{minimize}:

$$Q = (y_1 - \theta_1 - \theta_2)^2 + (y_2 - \theta_1 + 2\theta_2)^2 + (y_3 - 2\theta_1 + \theta_2)^2$$

For ~~this~~ [we note that given question matches the general assumptions of linear regression with ~~homoscedasticity~~ and $E(\epsilon_i) = 0$]

To minimize our ~~Q~~, we set:

$$\frac{\partial Q}{\partial \theta_1} = 0 \quad \text{and} \quad \frac{\partial Q}{\partial \theta_2} = 0$$

$$\frac{\partial Q}{\partial \theta_1} = -2(y_1 - \theta_1 - \theta_2) - 2(y_2 - \theta_1 + 2\theta_2) - 4(y_3 - 2\theta_1 + \theta_2) = 0$$

— (1)

and

$$\frac{\partial Q}{\partial \theta_2} = -2(y_1 - \theta_1 - \theta_2) + 4(y_2 - \theta_1 + 2\theta_2) + 2(y_3 - 2\theta_1 + \theta_2)$$

= 0

— (2)

This gives us:

$$-2y_1 + 2\theta_1 + 2\theta_2 - 2y_2 + 2\theta_1 - 4\theta_2 - 4y_3 + 8\theta_1 - 4\theta_2 = 0$$

and

$$-2y_1 + 2\theta_1 + 2\theta_2 + 4y_2 - 4\theta_1 + 8\theta_2 + 2y_3 - 4\theta_1 + 2\theta_2 = 0$$

which gives us:

$$12\theta_1 - 6\theta_2 = 2y_1 + 2y_2 + 4y_3 \quad \text{--- (3)}$$

and

$$-6\theta_1 + 12\theta_2 = 2y_1 - 4y_2 - 2y_3 \quad \text{--- (4)}$$

Now add ~~(3)~~ $2 \times (3) + (4)$ to get:

$$18\theta_1 = 6y_1 + 6y_3 \Rightarrow \theta_1 = \frac{y_1 + y_3}{3}$$

Add ~~(3)~~ $(3) + 2 \times (4)$ to get:

$$18\theta_2 = 6y_1 - 6y_2 \Rightarrow \theta_2 = \frac{y_1 - y_2}{3}$$

$\therefore$ LSE of θ_1 is $\frac{y_1 + y_3}{3}$

and

LSE of θ_2 is $\frac{y_1 - y_2}{3}$

05

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.
- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

(a) First we find variance of a uniform random variable on $(0, 1)$.
 This is $\int_0^1 \left(x - \frac{1}{2}\right)^2 dx = \int_0^1 \left(x^2 - x + \frac{1}{4}\right) dx = \left[\frac{x^3}{3} - \frac{x^2}{2} + \frac{x}{4}\right]_0^1 = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}$

Now, U_1 and U_2 are independent. $\text{Var}(X_1) = \frac{1}{12} = \text{Var}(X_2)$

$$\text{Cor}(X_1, X_3) = \frac{1}{12} = \text{Cor}(X_2, X_3)$$

$$\text{Cor}(X_1, X_2) = 0$$

$$\text{Var}(X_3) = \frac{2}{12} = \frac{1}{6}$$

Then ρ is given as: $\rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}}$

$$\begin{bmatrix} 1 & 0 & \frac{\text{Cov}(1,2)}{\sqrt{\frac{1}{12} \times \frac{2}{12}}} \\ 0 & 1 & \frac{\text{Cov}(1,2)}{\sqrt{\frac{1}{12} \times \frac{2}{12}}} \\ \frac{1/12}{\sqrt{\frac{1}{12} \times \frac{2}{12}}} & \frac{1/12}{\sqrt{\frac{1}{12} \times \frac{2}{12}}} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix}$$

(b) we find the eigenvectors of the matrix:

First, eigenvalues:

$$\begin{vmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{vmatrix} = (1-\lambda)^3 - \frac{1}{2}(1-\lambda) - \frac{1}{2}(1-\lambda) \Rightarrow \text{set to zero}$$

i.e. $(1-\lambda)^3 = 1-\lambda \Rightarrow \lambda = 0, 1, 2$ are the eigenvalues.

Then, the largest eigenvalue is $\lambda_1 = 2$.

How??

Let us find the corresponding eigenvector e_1 : take as $[v_1, v_2, v_3]$

$$\begin{bmatrix} 1-2 & 0 & 1/\sqrt{2} \\ 0 & 1-2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} -1 & 0 & 1/\sqrt{2} \\ 0 & -1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

??

Then : $-v_1 + \frac{v_3}{\sqrt{2}} = 0$ and $-v_2 + \frac{v_3}{\sqrt{2}} = 0$

which gives us $\left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1\right]^T$ which we can normalize to : $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right]^T$

$\therefore e_1' = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right]$

Then, the first principal ~~value~~ component is:

$$e_1'X = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right] \begin{bmatrix} v_1 \\ v_2 \\ v_1 + v_2 \end{bmatrix} = \frac{v_1}{2} + \frac{v_2}{2} + \frac{v_1 + v_2}{\sqrt{2}}$$

$$e_1'X = v_1\left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right) + v_2\left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right) = \left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right)(v_1 + v_2)$$

which will have variance as:

$$\text{Var}(e_1'X) = \left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right)^2 [\text{Var}(v_1) + \text{Var}(v_2) + 0] = \boxed{\left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right)^2 \times \frac{1}{12}}$$

which comes out as 0.12143

$\therefore$ 0.12143 is the variance of the first principal component

01

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

(a)

i. Complete linkage hierarchical procedure. Here we choose $d_{u,v} = \max_{i \in u, j \in v} d_{ij}$

Start:

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

here, closest are 1 and 2, with distance 1

∴ we group these 2

and update $d_{(12),3} = \max(d_{13}, d_{23}) = 11$

$d_{(12),4} = \max(d_{14}, d_{24}) = 5$

Then:

①

	(12)	3	4
(12)	0		
3	11	0	
4	5	3.5	0

here, closest are 3 and 4 with distance 3.5

then:

②

	(12)	(34)
(12)	0	
(34)	11	0

with distance

$d_{(12),(34)}$ as max of distances across

clusters which was between 1 and 3, i.e. 11

And finally, they all get clustered together as (1234), distance was 11

ii. Average linking procedure:

Here, on each update, the distance will be the sum of each pair across clusters divided by no. of pairs, i.e. for clusters U and V:

$$\frac{\sum_{i=1}^{N(U)} \sum_{j=1}^{N(V)} d_{u_i, v_j}}{N(U) N(V)}$$

∴ Start:

$$\begin{matrix} & 1 & 2 & 3 & 4 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 \\ 1 \\ 1 \\ 5 \end{bmatrix} & \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 3.5 \end{bmatrix} \end{matrix}$$

closest are 1 and 2, with distance 1.

Then:

$$\begin{matrix} & (12) & 3 & 4 \\ \begin{matrix} (12) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 \\ \frac{1+2}{2} \\ \frac{5+3}{2} \end{bmatrix} & \begin{bmatrix} 0 \\ 3.5 \end{bmatrix} & 0 \end{matrix} = \begin{matrix} & (12) & (3) & (4) \\ \begin{matrix} (12) \\ (3) \\ (4) \end{matrix} & \begin{bmatrix} 0 \\ 6.5 \\ 4 \end{bmatrix} & \begin{bmatrix} 0 \\ 3.5 \end{bmatrix} & 0 \end{matrix}$$

Here, closest are (3) and (4) with distance 3.5

Then:

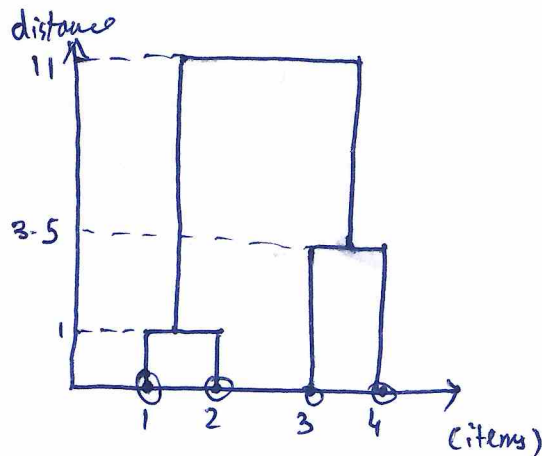
$$\begin{matrix} & (12) & (34) \\ \begin{matrix} (12) \\ (34) \end{matrix} & \begin{bmatrix} 0 \\ \frac{1+5+2+3}{4} \end{bmatrix} & 0 \end{matrix} = \begin{matrix} & (12) & (34) \\ \begin{matrix} (12) \\ (34) \end{matrix} & \begin{bmatrix} 0 \\ 5.25 \end{bmatrix} & 0 \end{matrix}$$

and finally, they all get clustered together as (1234) with distance

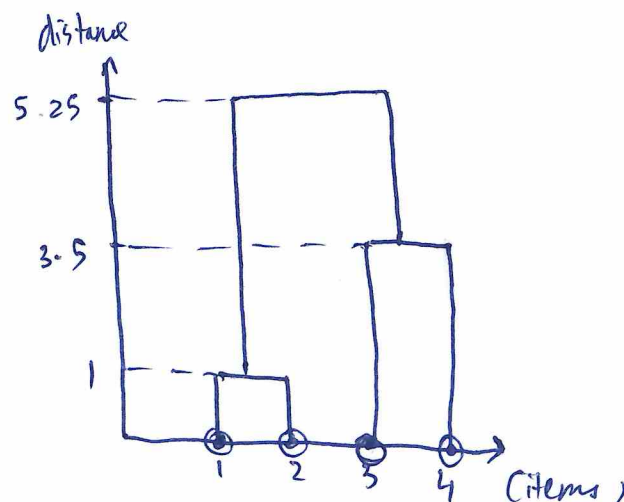
5.25.

(b) Dendrograms: i. ~~Maximum Linkage~~

i. Complete Linkage:



ii Average Linkage:



[continued on additional page, Page 8]

ADDITIONAL PAGES FOR ANSWERS

3(b) contd: We observe that for the given case both complete linkage and ~~can~~ average linkage give the same clusters. The distance between the clusters (1,2) and (3,4) however is different due to the different type of linkage, 11 for complete linkage and 5.25 for the average linkage.

03

